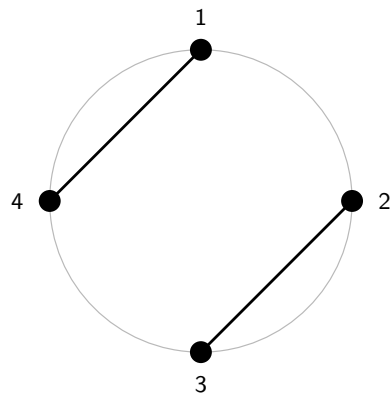
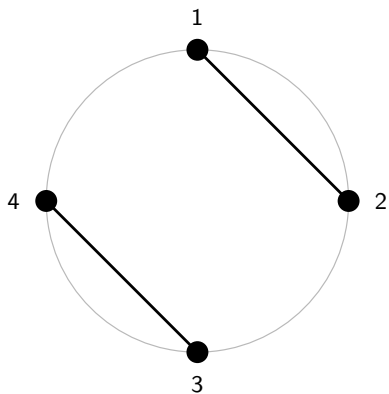


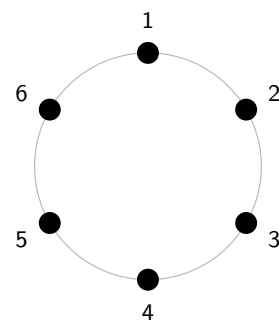
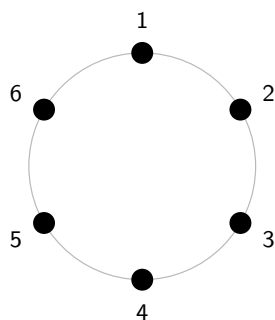
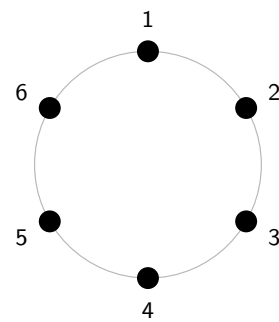
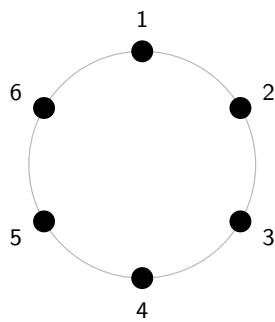
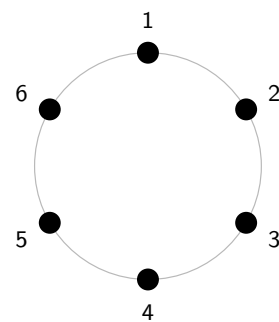
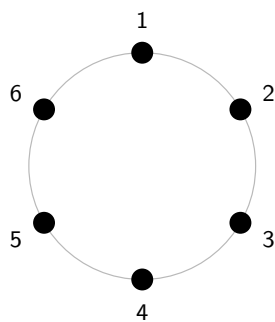
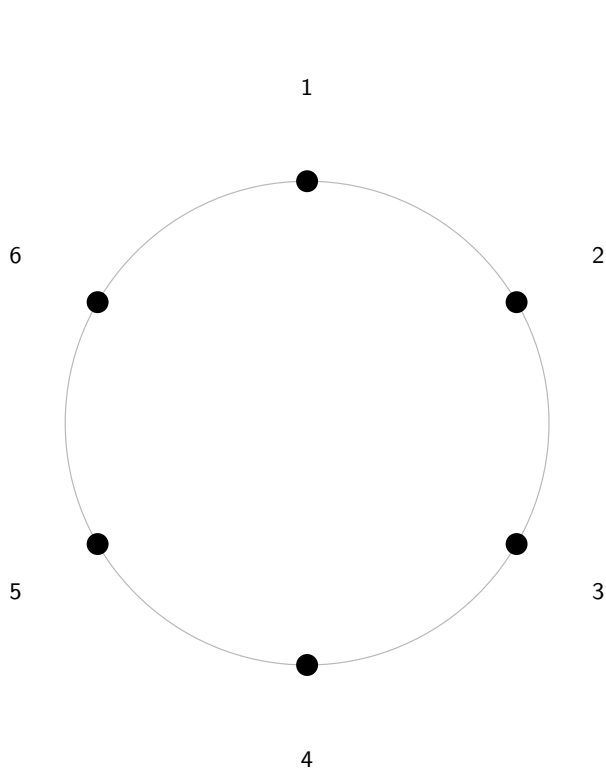
Week 12 / Catalan bijections / K-1

Use each dot once. Keep dots in place. Strings stay inside the circle without crossings. Pairings are different when a dot has a different partner. Use strings on large circles and a pencil on small circles.

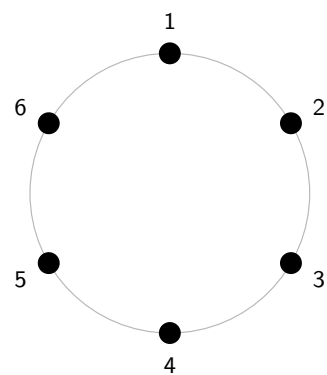
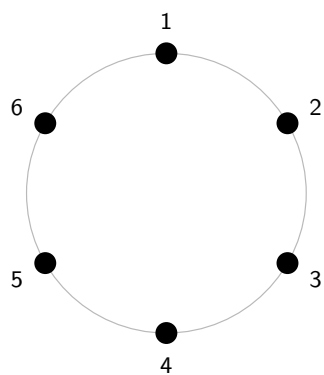
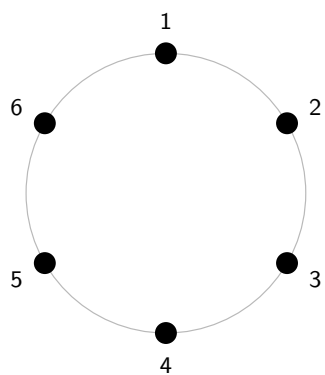
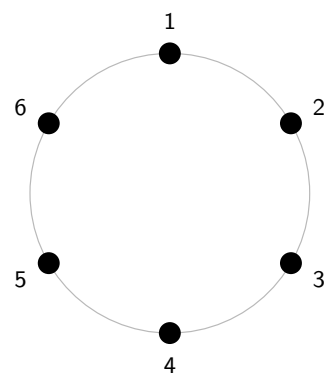
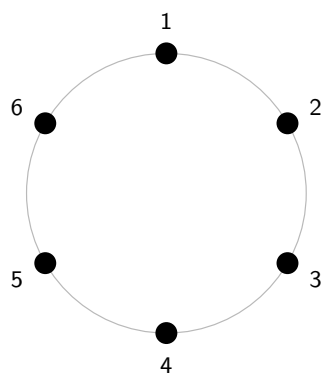
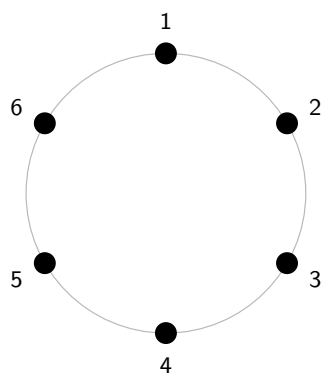
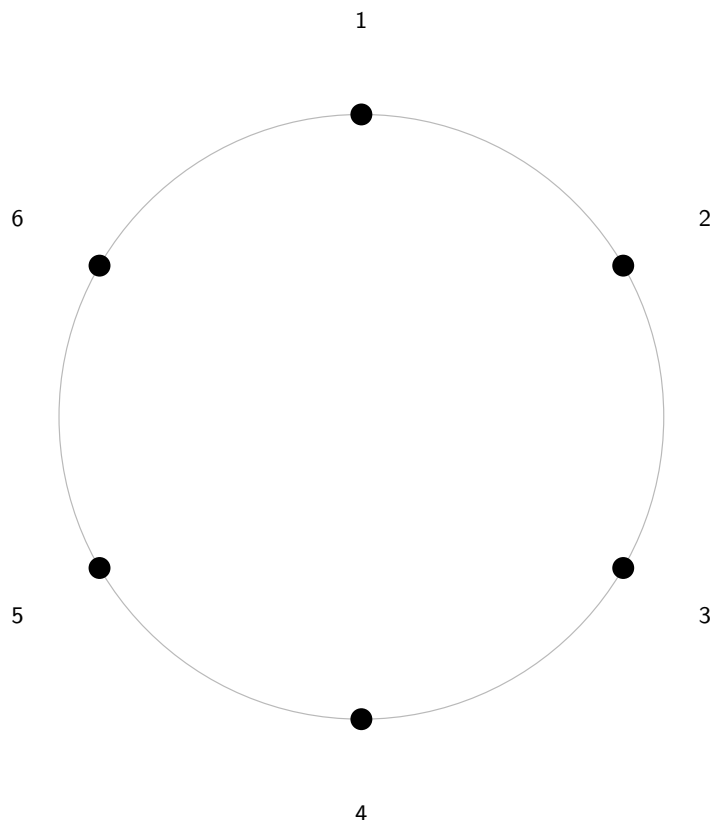
The two pairings of four dots:



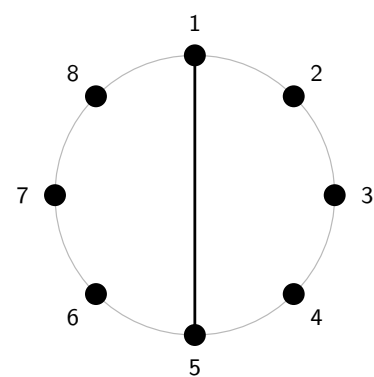
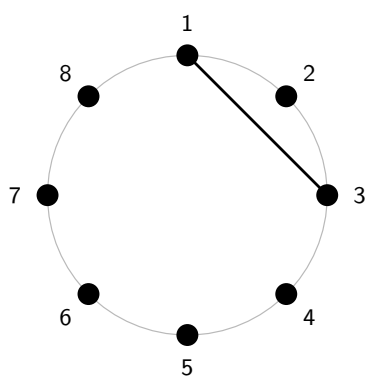
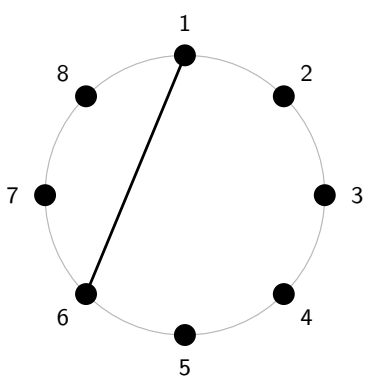
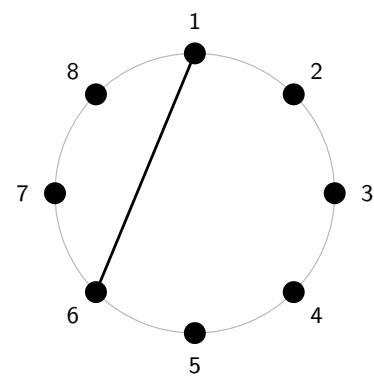
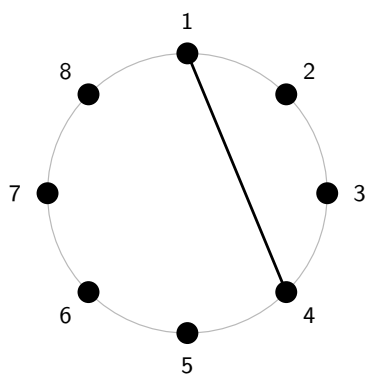
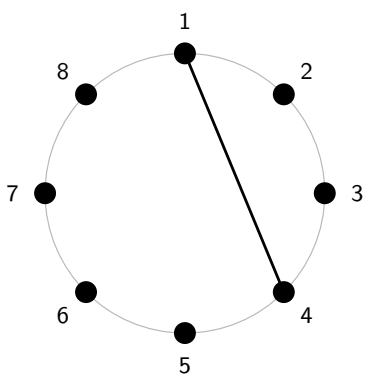
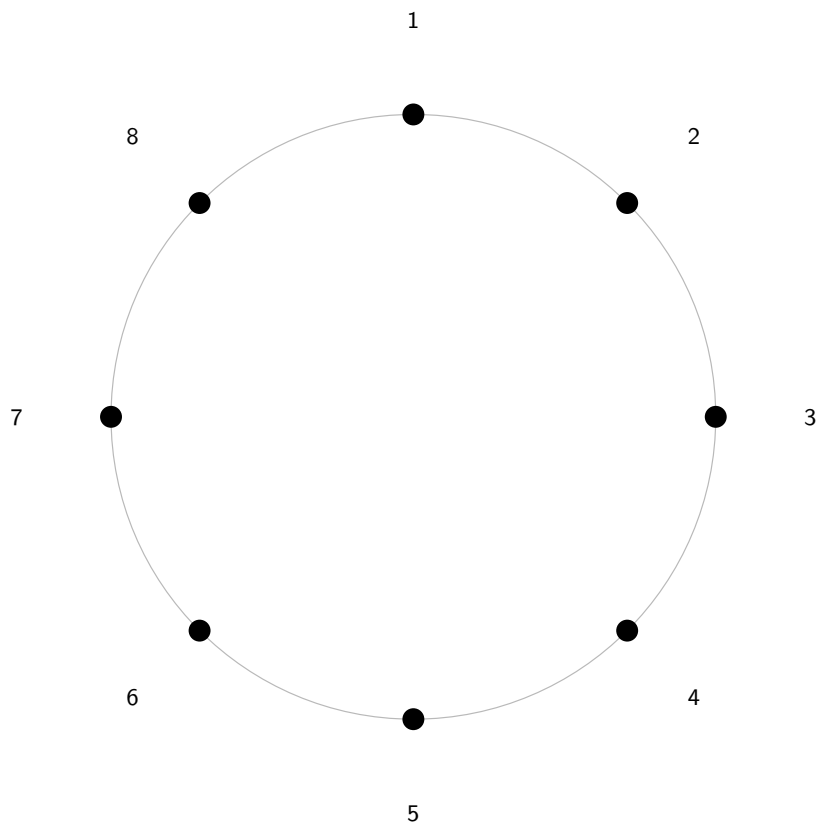
Problem 1: Find every way to pair six dots.



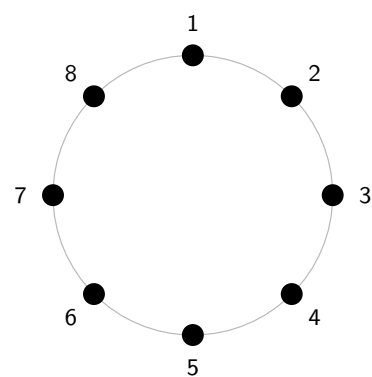
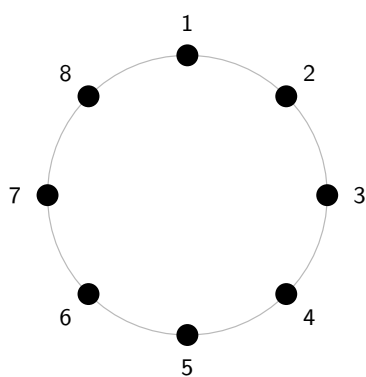
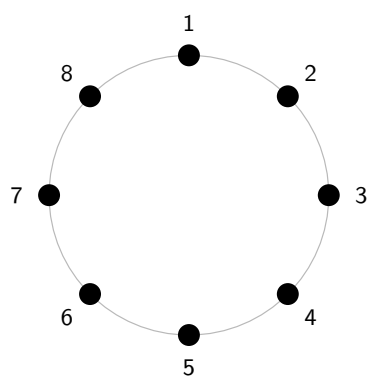
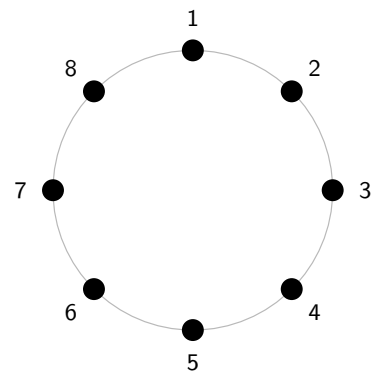
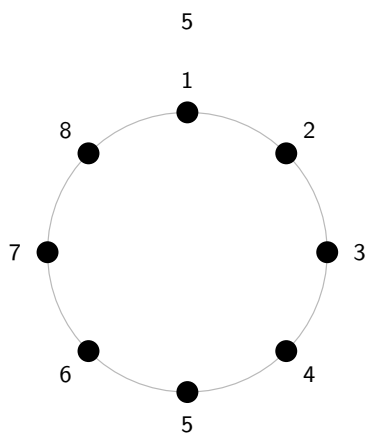
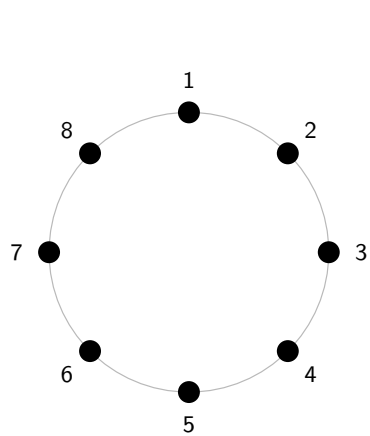
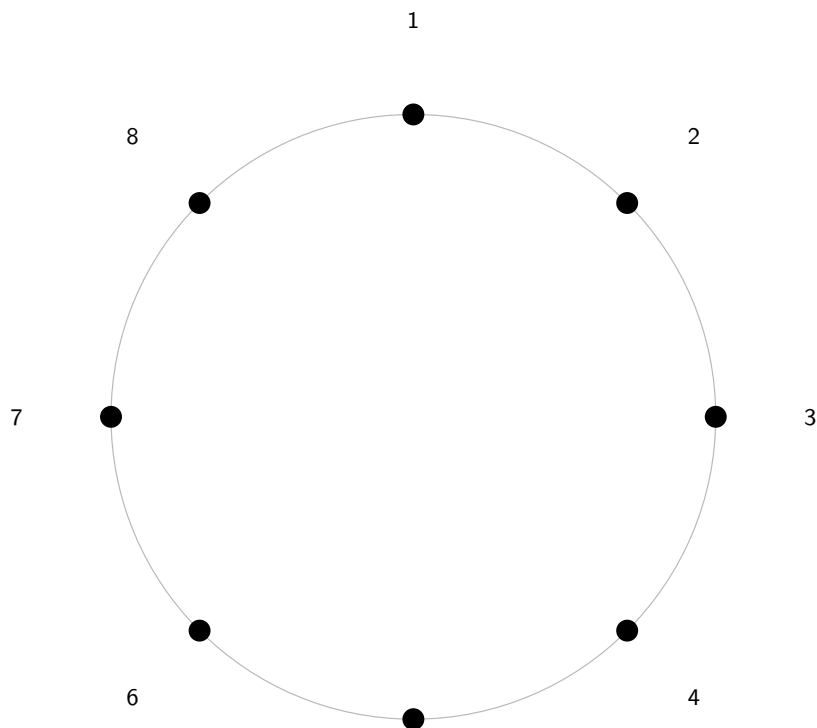
Problem 2: Check your six-dot collection. Show how you know that no pairing is missing.



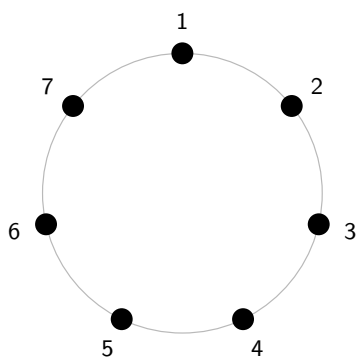
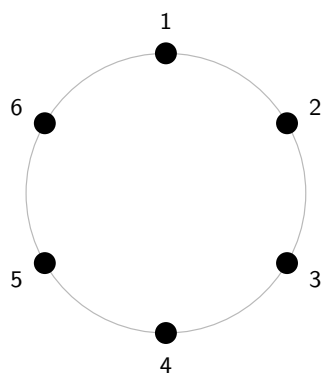
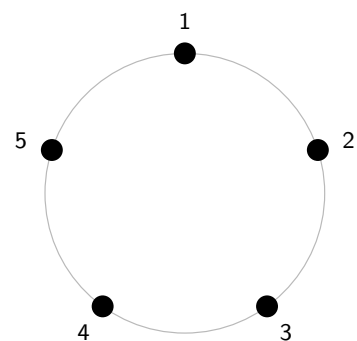
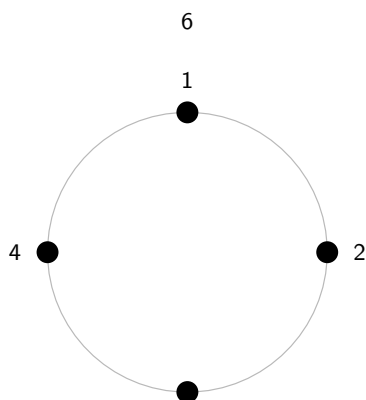
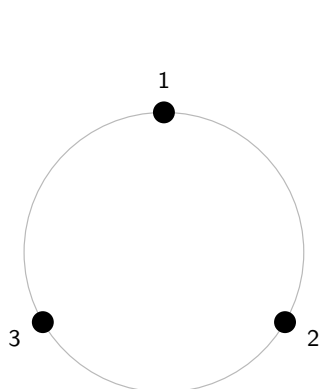
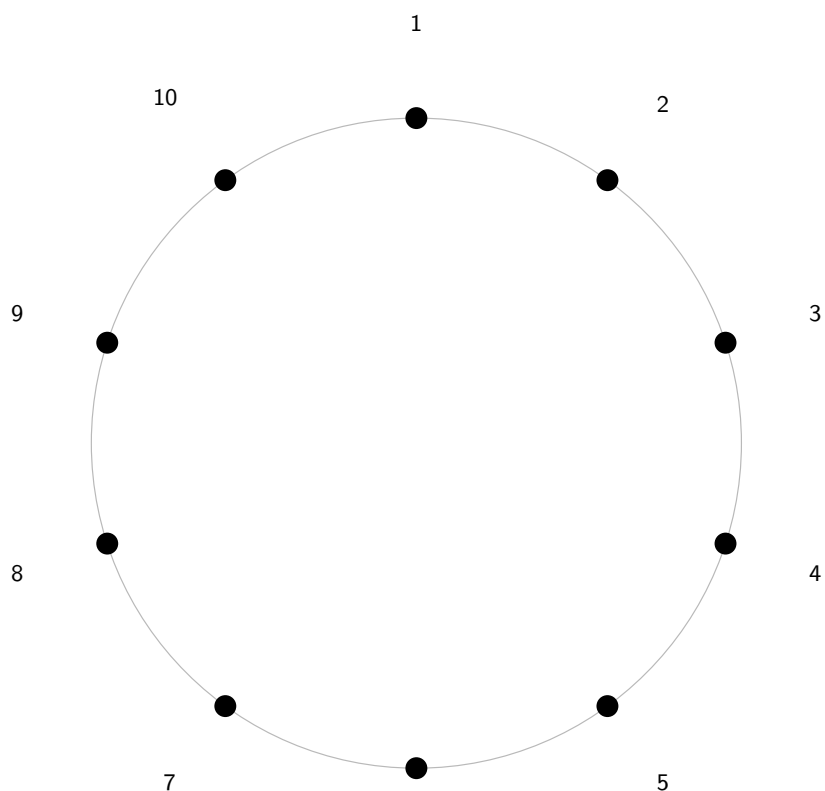
Problem 3: Keep each drawn pair when you try it on the large circle. Find every way to finish it, or show why you cannot.



Problem 4: Pair eight dots in as many different ways as you can. Draw extra circles on blank paper if you need them.



Problem 5: Pair all the dots in each circle if you can. If you cannot, show what gets in the way.



Problem 6: Can you pair eight dots without joining any two neighbors around the circle? Show why your answer is right.

